CONCEPT NOTE | DEEP LEARNING INDABAX BURUNDI 2026

Imbuto AI

A Kirundi-first, voice-enabled farm intelligence and early-warning platform for Burundi's smallholder farmers

Summary

Burundi's economy rests on smallholder, rain-fed farming, yet the people who grow the country's food have the least access to timely, usable information. Fewer than one in eight Burundians uses the internet, LTE reaches only a small share of the territory, and most digital tools assume literacy in French or English, a smartphone and a steady data connection. Meanwhile, delayed rains, record Lake Tanganyika flooding, high food prices and new refugee arrivals keep pushing households into crisis.

Imbuto AI (*imbuto*: seed) closes that gap. It combines a national agri-climate data platform with AI models that speak Kirundi, recognise crop disease from a photo, and forecast season shortfalls and flood risk. Farmers reach it by a simple voice call, SMS or USSD on any phone, or through an offline-first Android app; ministries, cooperatives and NGOs use a web dashboard. The project directly serves the agriculture pillar of Burundi's newly validated National AI Strategy.

> **The ask at IndabaX Burundi:** feedback from the Burundian ML community, introductions to data holders (IGEBU, Ministry of Agriculture, University of Burundi) and field partners, and Kirundi-speaking collaborators to co-build the speech and language components as open resources.

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1. The landscape: where the pain is

Our review of current public data points to four connected problems. They are shown below with the evidence behind each.

Problem	Evidence (latest available)	Why it matters for a solution
Food insecurity driven by climate shocks	About 1.77 million people (14% of the analysed population) were in IPC Phase 3 or worse in April-May 2026, including about 220,000 in Phase 4 (Emergency). Season 2026A harvests were below average after late, deficient rains. Around 86% of ~12.3 million people practise smallholder farming.	Farmers need season-specific, local guidance *before* they plant, not after the harvest fails.
Very low connectivity	Internet penetration was about 11% at end-2025 (DataReportal); mobile connections reach roughly 61% of the population. Most subscribers are still on GSM, and LTE covers under 16% of the country.	A smartphone-only app would miss most farmers. Voice, SMS and USSD must be first-class channels.
Language and literacy barrier	Kirundi is spoken by 9+ million people but is classed as a “scraping-by” language in NLP, with only a handful of open datasets (e.g. KIRNEWS) and early community efforts.	Advice delivered in Kirundi, by voice, is the difference between information and impact.
Thin extension reach and fragmented data	Field programmes report that farmers are not reached enough by extension workers. Rainfall, lake-level, price and food-security data sit in separate agencies and reports.	No single, trusted, machine-readable view exists to drive targeted advice or early warning.

Compounding pressures

- **Floods and displacement.** Lake Tanganyika reached record levels in 2024, displacing well over 100,000 people around Gatumba and Bujumbura. Burundi ranks among the 20 countries most vulnerable to climate change.
- **Economic strain.** Scarce foreign currency, a weakening franc and a roughly 50% rise in fuel prices push up food and transport costs.
- **Population movement.** About 230,000 Congolese refugees were in Burundi by February 2026, and Burundian returnees from Tanzania add pressure on host communities in the east.
- **Energy.** Only around one in ten people has electricity, so solutions must be light on power and data.

Why now

Burundi validated its National AI Strategy in April 2026 with UNDP support, naming agriculture among its priority sectors and planning incubators and financing for AI start-ups. A National Data Governance Strategy (November 2025), a 2026 personal data protection law, the BurundiPay instant-payment system (April 2026) and World Bank-funded rural 4G expansion are laying the rails. The policy window is open; what is missing are working, locally built applications.

2. The solution: Imbuto AI

Imbuto AI is designed around one principle: **meet the farmer where they are** — in Kirundi, on whatever phone they have, with or without a data connection. It has three parts.

2.1 Agri-Climate Lakehouse (data engineering)

A national, open-by-default data platform that ingests satellite rainfall and vegetation indices, weather and lake-level observations, IPC and market-price data, and farmer-reported observations. Pipelines clean and join these into a commune-level *gold layer* that every model and dashboard reads from. This is useful on its own: it gives government and partners a single, reproducible view of season conditions.

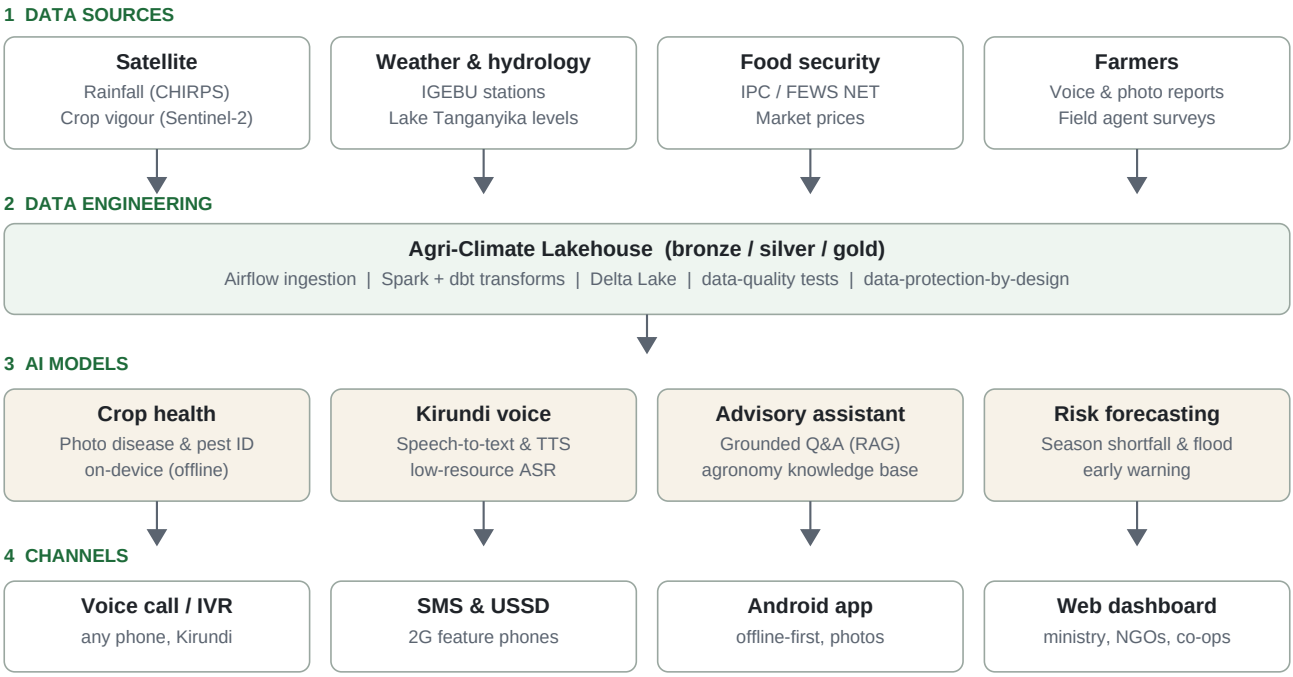
2.2 AI models

Model	What it does	Approach
Crop health	Identifies common diseases and pests in beans, maize, banana, cassava and coffee from a phone photo.	Compact vision model (e.g. MobileNet/EfficientNet-Lite) fine-tuned on public and locally collected images; runs on-device.
Kirundi voice	Turns a farmer's spoken question into text and reads answers back aloud.	Fine-tune open multilingual speech models; transfer learning from Kinyarwanda, which is mutually intelligible with Kirundi.
Advisory assistant	Answers questions on planting, inputs, storage and markets, grounded in vetted agronomy content.	Retrieval-augmented generation over an approved knowledge base; answers cite their source and defer to agronomists when unsure.
Risk forecasting	Flags likely season shortfalls and flood-exposed communes weeks ahead.	Gradient-boosted and time-series models on rainfall anomalies, NDVI and lake levels; validated against past IPC outcomes.

2.3 Channels

- **Voice call (IVR):** the default for any phone and any literacy level.
- **SMS and USSD:** planting-window alerts, flood warnings and price updates on 2G feature phones.
- **Android app:** offline-first, with photo diagnosis on the device and sync when a signal is available.
- **Web dashboard:** commune-level risk maps and aggregated farmer reports for ministries, cooperatives and NGOs.

3. System architecture



Farmer reports and outcomes flow back into the lakehouse to retrain models (feedback loop)

Proposed stack: Python, Apache Airflow, Spark and dbt on Delta Lake; MLflow for experiment tracking and model registry; FastAPI services; React web dashboard; Android client with TensorFlow Lite. Cloud-agnostic and containerised (Docker), with an option to host on in-country infrastructure as data-sovereignty rules require.

Design choices that fit Burundi

- **Low-bandwidth by default:** SMS/USSD and voice need no mobile data; the app works offline.
- **Human in the loop:** agronomists approve advisory content; the assistant escalates when confidence is low.
- **Privacy by design:** consent-based collection, minimal personal data, and alignment with Burundi's 2026 data protection law and National Data Governance Strategy.
- **Open resources:** Kirundi speech and agricultural text datasets released openly to grow the local NLP ecosystem, crediting contributors.

4. Roadmap

Phase	Timeline	Deliverables
0. Discovery	Months 0-2	Partner agreements; data audit; farmer interviews in 2 provinces; Kirundi data-collection protocol.
1. MVP	Months 2-6	Lakehouse v1; SMS/USSD alerts; crop-health model for 2 crops; dashboard prototype; Kirundi speech baseline.
2. Pilot	Months 6-12	Pilot with ~2,000 farmers via a cooperative or field partner; voice line live; app beta; impact baseline.
3. Scale	Months 12-24	More crops and provinces; flood early-warning with local authorities; open datasets and model release.

5. Impact and how we will measure it

Indicator	Pilot target (12 months)
Farmers actively using any channel (monthly)	2,000+, at least 40% women
Share of users on voice/SMS/USSD (non-smartphone)	60%+
Kirundi speech recognition word error rate	Measured baseline, then a 30% relative reduction
Crop-disease identification accuracy on local test set	85%+ top-1 for priority diseases
Lead time of shortfall/flood alerts	2+ weeks ahead of IPC or flood events in back-tests
Self-reported adoption of at least one recommended practice	50% of active users
Open datasets released	Kirundi speech + agri-text corpus, with documentation

Alignment

- **National AI Strategy 2025-2030:** applied AI in agriculture, local skills and data infrastructure.
- **SDGs:** 2 (Zero Hunger), 13 (Climate Action), 9 (Industry, Innovation and Infrastructure).
- **Deep Learning Indaba mission:** African-built ML on African data, in African languages.

6. Partners we seek

Partner type	Role
Ministry of Agriculture; IGEBU (geographic and meteorological institute)	Data access, content validation, alert dissemination
University of Burundi and local AI community	Kirundi language expertise, student research, evaluation
Field programmes and cooperatives (e.g. One Acre Fund/Tubura, AUXFIN G50 groups)	Pilot farmers, field feedback, distribution
Mobile network operators	Toll-free voice/USSD short codes, zero-rated access
UNDP, FAO, WFP and development funders	Co-funding, link to early-warning and humanitarian systems

7. Risks and mitigations

Risk	Mitigation
Too little labelled Kirundi speech	Community data drives with consent and fair pay; transfer from Kinyarwanda; start with a narrow agricultural vocabulary.
Wrong advice causing harm	Vetted knowledge base only; agronomist review; confidence thresholds and escalation to human advisers.
Data-sharing hesitancy from agencies	Start with open satellite data; show value early; formal agreements under the data governance framework.
Low trust or adoption	Co-design with farmer groups; deliver through trusted cooperatives; free core service.
Sustainability after pilot	Paid dashboard and analytics for NGOs, insurers and agribusiness; public co-funding for the free farmer tier.

8. Lead

Desmond Onam — Senior Instructor and consultant in Machine Learning, Data Engineering and Data Science, based in Kenya. Has trained 900+ professionals across Africa, Asia and the Middle East; built a large language model adopted by the Government of Bhutan and a data pipeline presented at the United Nations; designed a skills-to-role taxonomy for a Qatar workforce initiative. Works across Python, FastAPI, React, Spark, Kafka, Airflow, dbt, Delta Lake, MLflow, LangChain and AWS/GCP.

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References

[1] IPC / Knowledge4Policy (EC JRC). Burundi: Acute Food Insecurity Situation April-May 2026 and Projection for June-September 2026.

[2] FEWS NET. Burundi Food Security Outlook, February-September 2026; Outlook Update, December 2025.

[3] FAO GIEWS. Country Brief: Burundi, reference date 5 February 2026.

[4] DataReportal (Kepios). Digital 2026: Burundi.

[5] BuddeComm. Burundi Telecoms, Mobile and Broadband - Statistics and Analyses, 2026.

[6] Niyongabo, Qu, Kreutzer & Huang (2020). KINNEWS and KIRNEWS: Benchmarking Cross-Lingual Text Classification for Kinyarwanda and Kirundi. COLING.

[7] Netherlands Space Office (G4AW). GAP4A: Good Agricultural Practices for All - digitizing agriculture in Burundi, 2018-2021.

[8] DiploFoundation Digital Watch. Burundi country profile (National AI Strategy, data governance, BurundiPay), 2026.

[9] WeAreTech Africa / OSIRIS. Burundi validates its National AI Strategy, 23 April 2026; Burundi presents AI ambitions in Seoul, 10 September 2026.

[10] African Arguments; Al Jazeera. Reporting on Lake Tanganyika record levels and flooding in Burundi, 2024-2025.

[11] Lighting Global / World Bank. Burundi off-grid market and electricity access.